ESP32 IoT Sensor Controller — Bill of Materials

Project: ESP32_IoT_Sensor_Controller
Design Tool: KiCad 10 | PCB: 2-Layer
Status: Design completed; manufacturing and physical testing pending

Total quantity / board **36**

Reference	Qty	Value	Footprint	Datasheet	Manufacturer	MPN
C1,C4,C8,C9	4	100nF	Capacitor_SMD:C_0603_1608Metric	Datasheet	TDK	C1608X7R1H104K080AA
C2,C5,C6,C7,C10	5	10uF	Capacitor_SMD:C_1206_3216Metric	Datasheet	Samsung Electro-Mechanics	CL31B106KAHNNNE
C3	1	1uF	Capacitor_SMD:C_0603_1608Metric	Datasheet	TDK	C1608X5R1A105K080AC
D1	1	LED Green	LED_SMD:LED_0603_1608Metric	Datasheet	Kingbright	APT1608SGC
D2	1	SMBJ5.0A	Diode_SMD:D_SMB	Datasheet	STMicroelectronics	SMBJ5.0A-TR
D3	1	USBLC6-2SC6	Package_TO_SOT_SMD:SOT-23-6	Datasheet	STMicroelectronics	USBLC6-2SC6
F1	1	PTC 1.1A 16V	Fuse:Fuse_1206_3216Metric	Datasheet	Littelfuse	1206L110/16WR
J1	1	USB-C TYPE-C-31-M-12	ESP32_IoT_Custom:USB_C_HRO_Custom	Datasheet	HRO	TYPE-C-31-M-12
J2	1	External 3V3 / GPIO	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54 mm_Vertical	Datasheet	Samtec	TSW-104-07-G-S
Q1,Q2	2	MMBT3904	Package_TO_SOT_SMD:SOT-23	Datasheet	onsemi	MMBT3904LT1G
R1,R2	2	5.1k	Resistor_SMD:R_0603_1608Metric	Datasheet	YAGEO	RC0603FR-075K1L
R3,R4,R11,R12	4	10k	Resistor_SMD:R_0603_1608Metric	Datasheet	YAGEO	RC0603FR-0710KL
R5,R6	2	4.7k	Resistor_SMD:R_0603_1608Metric	Datasheet	YAGEO	RC0603FR-074K7L
R7,R10	2	1k	Resistor_SMD:R_0603_1608Metric	Datasheet	YAGEO	RC0603FR-071KL
R8	1	22.1k	Resistor_SMD:R_0603_1608Metric	Datasheet	YAGEO	RC0603FR-0722K1L
R9	1	47.5k	Resistor_SMD:R_0603_1608Metric	Datasheet	YAGEO	RC0603FR-0747K5L
SW1,SW2	2	B3F-1000	Button_Switch_THT:SW_PUSH_6mm	Datasheet	Omron	B3F-1000
U1	1	ESP32-WROOM-32E	ESP32_IoT_Custom:ESP32-WROOM-32E_Custom	Datasheet	Espressif	ESP32-WROOM-32E-N4
U2	1	CP2102N-A02-GQFN24	Controller:QFN24_4x4mm_P0.5mm_CP2102N	Datasheet	Silicon Labs	CP2102N-A02-GQFN24R
U3	1	AP7361C-33FGE-7	Controller:AP7361C_UDFN3030_8E	Datasheet	Diodes Incorporated	AP7361C-33FGE-7
U4	1	SHT31-DIS	Sensor_Humidity:Sensirion_DFN-8-1EP_2.5x2.5mm_P0.5mm_EP1.1x1.7mm	Datasheet	Sensirion	SHT31-DIS-B